Group Coursework: Forecasting Oil Prices

**Autumn 2025 Financial Econometrics**

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- **Written Report (max 8 pages – approx.. 2,000 words):** Wednesday Week 7 (report and program files) – 25%

**1. Coursework Overview**

Imagine you are a junior analyst working for an oil trading desk. Your task is to develop **a forecasting model** that can help a senior trader form expectations about oil futures returns and identify potential trading opportunities.

The trader is not an econometrician. They care about whether your model produces useful, **interpretable signals** that can inform trading decisions. They also want to know what data and reasoning underpin your forecasts, how reliable the model is, and what risks or limitations they should be aware of.

Your role is therefore twofold:

1. Technical – Apply econometric methods to forecast oil futures, carefully selecting the dependent variable (e.g. CL1 price, CL1 returns, or the CL2–CL1 spread), the relevant predictors (from the macroeconomic and industry dataset provided), and the appropriate data frequency. Evaluate your model with diagnostic tests and out-of-sample performance checks.
2. Communicative – Translate your results into clear economic and trading insights that a busy oil trader can understand and act on.

Deliverables:

- A written report that balances econometric rigour with trader-oriented communication.
    - Limit: 8 pages excluding the title page, bibliography and appendices.
    - The main text should include the abstract as well as the main tables and figures.
    - This corresponds to approximately 2,000 words.
    - The appendix can be used for secondary results, like full tables of diagnostic tests and extensions that are not core to the results.
- Two files containing your code and raw results
    - A jupyter notebook (.iupnb)
    - An exported version of the jupyter notebook (.pdf or .html) containing the results.

**2. Data**

The dataset (*Oil Data.xlsx*) contains:

- CL1: front-month crude oil futures price.
- CL2: second-month crude oil futures price.
- Additional macroeconomic and industry data (e.g. exchange rates, inventories, interest rates, global demand proxies).

It is your task to decide which variables are relevant for forecasting. Be explicit about how you deal with differences in frequency across variables.

**3. Methodology**

You are free to select your approach, provided you justify your choice. Techniques introduced in class include:

- Predictive regressions using OLS.
- ARIMA and ARIMAX models.
- LASSO and Ridge Regressions.

Some groups may be familiar with more advanced statistical techniques and keen to show them. You may use machine learning methods (e.g. random forests, neural networks), **BUT**: They are not the focus of this assignment! If you choose to include them, show how their results compare to more traditional econometric approaches. You will not be rewarded simply for using advanced tools — the emphasis is on justification, transparency, and communication.

You are free to use any appropriate econometric software. The module and tutorials mostly cover R implementations, but you are welcome to use Python or Julia. If you want to use other software, such as Matlab, SAS, Eviews, Stata, or other statistical software you feel comfortable with, please discuss with me first.

**4. Grading Criteria**

Your grade will depend on:

- **Clarity of explanation** – Could a trader understand your reasoning?
- **Diagnostics** – Did you check whether your model satisfies basic econometric assumptions?
- **Interpretation** – Can you explain why the model performs as it does and what it means economically?

Your report **must**:

- Justify your choice of dependent variable, data frequency, and predictors.
- Interpret key diagnostics (e.g., stationarity, autocorrelation, heteroskedasticity, parameter stability).
- Evaluate forecast performance using out-of-sample metrics and compare your model with at least **one simple naïve benchmark**.
- Synthesise results into **economic and trading insights.**

A **strong** report will:

- Be clearly structured, with **abstract, introduction, methodology, results, and conclusion**.
- Frame the research question in **trader-relevant terms** (e.g., "How can we forecast oil returns to support trading decisions?").
- Explains data and frequency choices in a way that makes sense to a trader.
- Justify the **econometric methodology** and report diagnostic checks succinctly.
- Present results in a **clear, visual, and intuitive** way.
- Offer **thoughtful interpretation**: what do the results mean for an oil trader?
- Remain **accessible to a finance professional** with limited econometrics background.
- Outlines and evaluates a (simple) trading rule.

## Grading Rubric

| Element | | Methodology | Analysis, interpretation | Structure | Presentation | Overall |
|---|---|---|---|---|---|---|
| Weig | Descriptor | 40% | 40% | 10% | 10% | 100% |
| 80+ | Outstanding (High Distinction) | Your work demonstrates exceptional knowledge of econometrics and the relevant forecasting techniques. The methodology is explained and applied expectionally well. | The paper displays abundant evidence of independent thought and a very strong critical analys of your results and reports them in a way that would be of very high value to a practioner. | Your work displays an excellent structure and very high ability to synthesise information to a very high level. | It is presented beautifully to a publication standard and fully adheres to academic best practice. | Overall, your work demonstrates exceptional skills in critical analysis and is of very high standard. |
| 70s | Excellent (Distinction) | Your work demonstrates extensive knowledge of econometrics and the relevant forecasting techniques. The methodology is explained very well and displays an extensive knowledge of the issues at hand. | The paper shows clear evidence of independent thought and contains critical analys of your results in a way that would be of high value to a practictioner. | Your work displays an very good structure and high ability to synthesise information to a high level. | It is presented nicely and generally adheres to academic best practice. | Overall, your work demonstrates extensive knowledge of the subject area and very good analysis. |
| 60s | Very Good (Merit) | Your work demonstrates an above average knowledge of econometrics and the relevant forecasting techniques. The methodology is described clearly and your modelling choices show an above average understanding of the material. | The paper shows evidence of independent thought and ability to analyse your results and their implications in a way that would be of considerable value to a practictioner. | Your arguments are sound and coherent and you show an ability to synthesize information to an above average level. | It is presented well, with most academic principles applied. | Overall, your work is clearly written and shows above average analysis. |
| 50s | Good (Pass) | Your work demonstrates some knowledge of econometrics and the relevant forecasting techniques but lacks in scope and/or clarity. You display knowledge of the models and techniques at hand. | The paper shows limited evidence of independent thought and limited critical analys of your results in a way that would be of some value to a practitioner. | Arguments are adequate, but there is limited synthesis of information. | It adheres to some of the academic best practice, but falls short in some areas. | Overall, your work is reasonable, but has significant scope for improvement. |
| <50 | Insuficient (Fail) | Your work demonstrates limited knowledge of of econometrics and the relevant forecasting techniques and significantly lacks in scope and clarity. The work demonstrates limited knowledge of the relavant models. | The paper shows minimal evidence of independent thought and no critical analys of your results and would be of minimal value to a practioner. | The arguments are poorly structured with little synthesis of information. | It is poorly presented and does not follow most academic best practices. | Overall, your work shows limited evidence of good analysis and has significant deficiencies. |

## 5. Hints and Guidance

- **Target Variable** – Will you forecast the level of CL1, its returns, or the CL2–CL1 spread? Which is most useful for a trader? What econometric properties do each choice have?
- **Frequency** – Would a trader care more about daily, weekly, or monthly forecasts? What are the pros and cons of each choice? Make sure to stick with one frequency once you have made your choice.
- **Predictors** – How do macroeconomic and industry variables help? Be selective and explain why you include or exclude them.
- **Depth vs. Breadth –** Avoid the trap of trying to do too much. After some initial exploration, hone in on one or two models that you want to estimate and focus your energy on explaining those clearly and convincingly to your target audience.
- **Model Comparison** – Use appropriate metrics to compare models. Out-of-sample performance matters more than in-sample fit.
- **Model Complexity** – A more complex model is not automatically better. You will be rewarded for selecting the right level of complexity for the problem, justifying your choice, and communicating the trade-off between interpretability and predictive power. A simple, well-motivated model explained clearly is often more valuable to a trader than a black-box approach.
- **Trading Angle** – If your forecasts show promise, outline how they could translate into a trading rule (while noting practical limitations).
- **Assessment Focus** – You are NOT graded on whether your forecasts "beat the market." You are graded on how well you justify your model, test it, and communicate its relevance to trading.